

Markscheme

Mathematics: analysis and approaches

Practice question bank

110. (a)

In the expansion of $(1 + x)^n$, three consecutive coefficients $\binom{n}{k-1}$, $\binom{n}{k}$, $\binom{n}{k+1}$ are in the ratio $1 : 2 : 3$.

Find the value of n and the value of k .

$$\binom{n}{k-1} : \binom{n}{k} = k : (n - k + 1) = 1 : 2 \Rightarrow n = 3k - 1 \quad (M1) \text{ A1}$$

$$\binom{n}{k} : \binom{n}{k+1} = (k + 1) : (n - k) = 2 : 3 \Rightarrow 2n = 5k + 3 \quad (M1) \text{ A1}$$

$$\text{substitute } n = 3k - 1: 2(3k - 1) = 5k + 3 \Rightarrow k = 5 \quad (M1)$$

$$n = 3(5) - 1 = 14 \quad \text{A1}$$

[7 marks]